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JUNE 10, 1959
SUPERSEDING
Fed. Spec. DDD–D–666
March 24, 1953

FEDERAL SPECIFICATION

PAD, COTTON, DRESSING, EYE PAD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers sterile dressings for use as a protective covering for the eye.

1.2 Classification.

1.2.1 *Types.* — The dressings covered by this specification shall be of the following types, as specified (see 6.1):

Type I. —Covered with gauze.
Type II.—Covered with nonwoven cotton fabric.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU–P–31 — Paper; General Specifications and Methods of Testing.
CCC–G–101—Gauze, Absorbent.
CCC–T–191—Textile Test Methods.
JJJ–C–561 — Cotton; Sterile and Nonsterile.
PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–636—Boxes, Fiber.
PPP–B–676 — Boxes, Set-Up, Paperboard.
PPP–T–0045 — Tape, Gummed, Paper, Reinforced.
PPP–T–76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking For Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.—The following publication forms a part of this specification.

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Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U. S. Pharmacopoeial Convention, Inc.:
Pharmacopoeia of the United States.

(Application for copies should be addressed to the Mack Printing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Absorbent cotton.*—The cotton shall meet the requirements of Federal Specification JJJ–C–561 for grade B cotton, when tested as described in 4.4.1.

3.1.2 *Nonwoven cotton fabric.*—The nonwoven cotton fabric shall consist of all cotton, or a mixture of cotton and rayon, and shall meet the following requirements when tested in accordance with 4.4.2.

3.1.2.1 *Submersion time.*—Not more than 30 seconds shall be required for complete submersion, when tested in accordance with 4.4.2.1.

3.1.2.2 *Water extract.*

3.1.2.2.1 *Residue weight before ignition.*—Residue weight before ignition shall be not more than 3.5 percent of the weight of the fabric, when tested in accordance with 4.4.2.2.1.

3.1.2.2.2 *Residue weight after ignition.*—Residue weight after ignition shall be not more than 0.5 percent of the weight of the fabric, when tested in accordance with 4.4.2.2.2.

3.1.2.2.3 *Acid or alkali.*—Test results shall not show a pink color, when tested in accordance with 4.4.2.2.3.

3.1.2.2.4 *Starch.* — Test results shall not show a red, violet, or blue color, when tested in accordance with 4.4.2.2.4.

3.1.2.3 *Fatty matter.*—Not more than 8 percent of residue shall remain when tested in accordance with 4.4.2.3.

3.1.2.4 *Ash content.*—The weight of the ash shall be not more than 0.85 percent of the weight of the fabric taken when tested in accordance with 4.4.2.4.

3.1.2.5 *Uncombined dyes.*—The percolate may show a yellowish color as a result of the test, but neither a blue nor a green tint shall be apparent, when tested in accordance with 4.4.2.5.

3.1.2.6 *Color.* — Color shall be white or cream.

3.1.3 *Gauze.*—The gauze shall meet the requirements of Federal Specification CCC–G–101 for type I, when tested in accordance with 4.4.3.

3.1.4 *Adhesive.* — The adhesive shall be white or colorless, sterilizable, and nonbleeding.

3.1.5 *Paper envelope.*—The paper envelope shall be of glassine paper weighing 25 pounds, basis 24 by 36 inches (500 sheets), when tested in accordance with 4.4.4.

3.2 Fabrication.

3.2.1 *Type I dressing.*—The type I dressing shall be of the shape and dimensions as shown in figure 1. The dressing shall be composed of a filler of absorbent cotton covered top and bottom with single thicknesses of gauze, all edges of the dressing being open. The weight of the finished dressing shall be 1.72 gram (g.) (plus or minus 0.26 g.).

3.2.2 *Type II dressing.*—The type II dressing shall be of the shape and dimensions shown in figure 2. The dressing shall be composed of a filler of absorbent cotton enclosed in nonwoven fabric to form a flattened tubular pad. The dressing shall be formed by wrapping a layer of absorbent cotton with a sheet of nonwoven fabric so that the short edges of the dressing are covered and the long edges open. The free ends of the covering shall overlap each other on the reverse side for $\frac{3}{8}$ inch (plus or minus $\frac{1}{8}$ inch) as shown in figure 2. The free ends shall be glued to each other. The weight of the finished dressing shall be 1.36 g. (plus or minus 0.16 g.).

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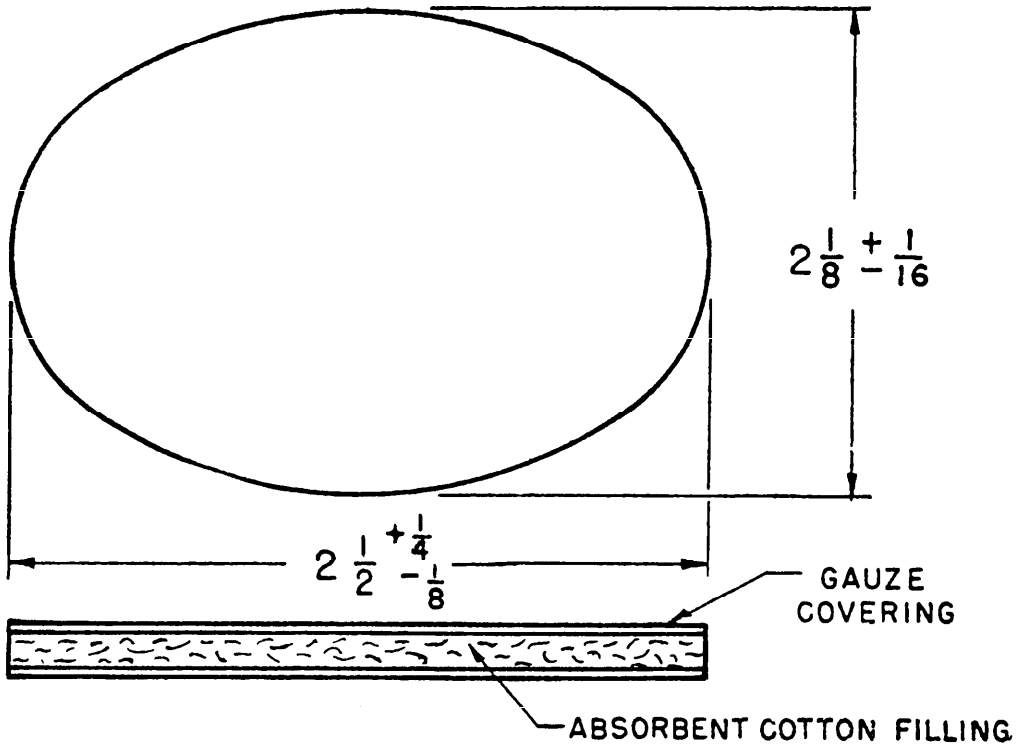


FIGURE 1.

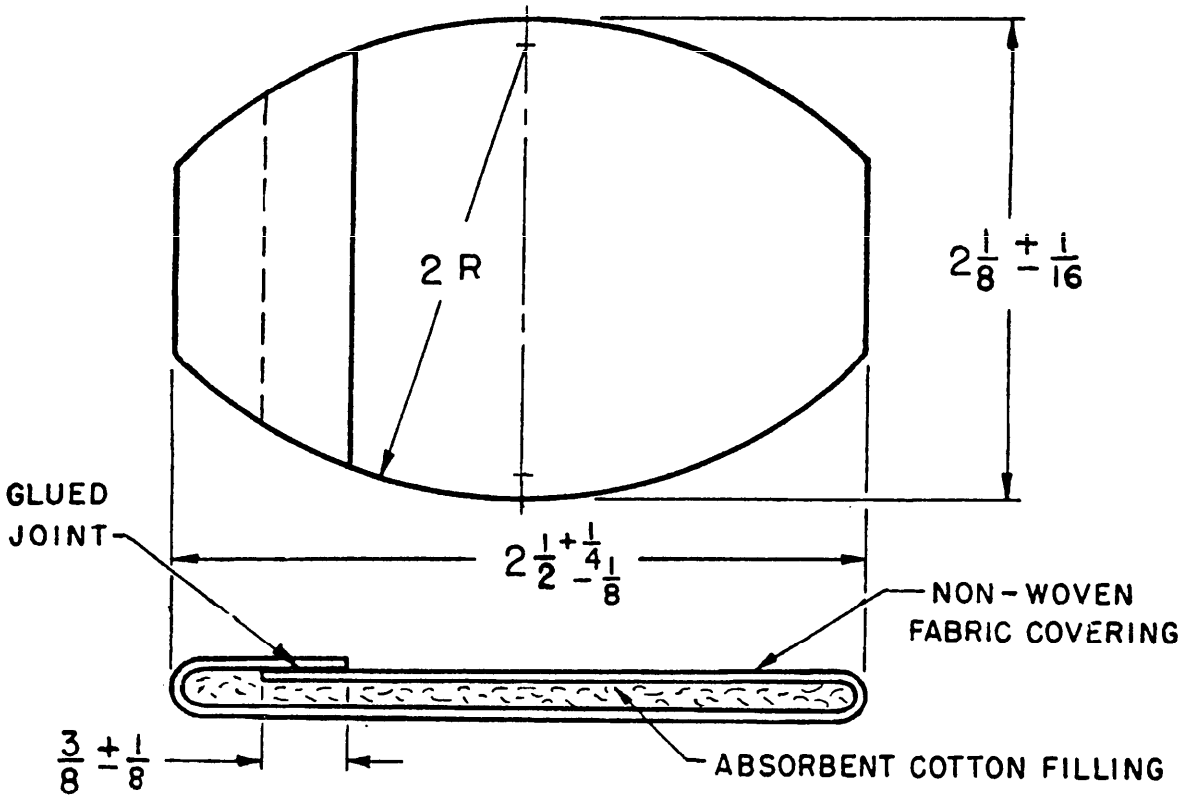


FIGURE 2.

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3.3 **Package.** — Each individual dressing shall be enclosed in a glassine paper envelope, as specified in 3.1.5, and sealed. The envelope shall be so constructed and sealed that sterility of the contents is maintained until the envelope is opened.

3.4 **Requirements for certificates.** — Compliance with (*a*) or (*b*) is required.

(*a*) *Certificate for steam (water vapor) sterilization.* — Actual accomplishment of sterilization for each sterilizer charge shall be certified in writing by sworn affidavit as having been accomplished by the manufacturer. Such certificate shall list the lot or control number of each sterilizer charge and shall state that the charge was subjected to saturated steam, at a minimum of 250°F., for not less than 20 minutes after sterilizing conditions have been attained (when gauge pressure corresponds to that of saturated steam at the indicated temperature). Gauge pressure readings shall also be reported. Said readings shall correspond to the required pressure for saturated steam and temperature in the autoclave. "Sterilization detectors", of the type indicating duration of time to steam and temperature, shall be used to indicate that the desired sterilization conditions have been attained. Detectors shall be placed in the center, as well as distributed longitudinally and transversely throughout the sterilizer load. The gauge pressure readings and the sterilization detectors shall be maintained as a permanent record by the contractor and shall be made available upon request.

Note.—Temperatures as low as 240°F. may be used, provided it is demonstrated that effective sterilization is obtained.

(*b*) *Certificate of sterility for steam (water vapor) or other methods of sterilization.*—A certificate of sterility, listing all lot or control numbers, stating that samples from each sterilizer charge have been tested and found to be sterile, shall be submitted to the procuring activity. This certificate shall include specific tests performed and the test results.

3.5 **Workmanship.**—Workmanship shall be first class throughout. Cotton eye pads shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The provisions contained in this section are applicable to inspection performed by and for the Government. Sampling, inspection, and test procedures for components or materials described shall be in accordance with this specification and with specifications and standards referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.1.1 *Supplier responsibilities for inspection.*—Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request at any time or from time to time for a period of 2 years after delivery of the supplies to which such records relate.

4.1.2 Inspection, as used in this specification, is defined as both examination (such as, visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 **Sampling.** — Sampling shall be conducted in accordance with Military Standard MIL–STD–105 and appendix thereto. Unless otherwise specified herein, the unit of product for inspection purposes shall be the individual pad.

Defects	Inspection level	AQL (percent defective)
Major A	L8	1.0
Total A	L8	4.0
Major B	L3	1.0
Total B	L3	4.0
Minor C	L8	2.5
(Unit of product — package)		

4.2.1 *For visual examination.*—Sampling procedures for visual examination, when in-process inspection can be performed, shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105. The unit of product for inspection purposes shall be the individual pad unless otherwise specified. Where in-process inspection cannot be performed, sample size of the end item shall be as indicated for inspection level L8 in appendix to Military Standard MIL–STD–105. Sample size for major B and minor B defects shall be as indicated for inspection level L3 in appendix to Military Standard MIL–STD–105.

4.2.2 *For tests.*—Sufficient samples of the components shall be selected to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by subsidiary specifications is not specified, the sample size for each test shall be as indicated for inspection level L4 in appendix to Military Standard MIL–STD–105. No test result shall fall below the minimum or exceed the maximum values specified. Not less than three samples shall be taken regardless of lot size. Lot sizes are expressed as follows:

Cotton _____pound.
Gauze _____pound.
Nonwoven fabric _____yard.

4.2.3 Where possible, the same sample shall be used for the determination of two or more tests.

4.3 **Inspection.**—The pads shall be inspected to determine compliance with all requirements contained in sections 3 and 5 of this specification. Nonconformance with these requirements will be permitted to the extent indicated in 4.2.

4.3.1 *Classification of defects.*—Inspection shall be conducted in accordance with the following classifications of defects:

*Categories and defects**

Major A:
101. Covering not free of holes, cuts, or tears.
102. Paper envelope not free from tears or other damages.

Major B:
151. Weight of finished pad not as specified.

Minor A:
201. Pad not uniformly assembled.
202. Pad not free from dirt, ravelings, or other foreign matter.
203. Type II pad not properly glued.

Minor B:
251. Dimensions not within specified tolerances.

Minor C:
301. Labeling or printing on package not legible and free from defacing marks, scratches, or tears.

* Inspection is not restricted to the classified possible defects listed above.

4.4 **Tests.**

4.4.1 *Absorbent cotton.*—Absorbent cotton shall be tested for compliance with Federal Specification JJJ–C–561.

4.4.2 *Nonwoven fabric.* — The nonwoven fabric shall be tested for compliance with the requirements described in 3.1.2 by test procedures specified hereinafter.

4.4.2.1 *Submersion time.*—A 5-g. sample of nonwoven fabric, torn into five equal pieces, shall be tested for submersion time using the

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equipment and method specified for absorbency of purified cotton, as described in the Pharmacopoeia of the United States (U.S.P.).

4.4.2.2 *Water extract.*—A 20-g. (plus or minus 100 mg.) specimen of the nonwoven fabric shall be placed in 500 cc. of distilled water and boiled for 15 minutes, boiling distilled water being added as necessary, to maintain the original volume. The water shall then be poured through a funnel into a 1000 cc. volumetric flask, the test specimen transferred to the funnel, the excess water pressed out of the test specimen with a glass rod, and the fabric washed with two successive 250 cc. portions of boiling distilled water (the fabric being pressed after each washing). The washings shall be combined to make the total volume 1000 cc.

4.4.2.2.1 *Residue weight before ignition.*—Of the water extract, 400 cc. shall be evaporated (filtered if necessary) in a suitable dish on a water bath and the residue dried to constant weight at 105° C. The weight of the dried residue shall be multiplied by 2.5. This product shall be defined as the weight of the residue before ignition.

4.4.2.2.2 *Residue weight after ignition.*—To determine residue weight after ignition, the above resultant residue (4.4.2.2.1) shall be ignited to constant weight in a muffle furnace at a dull-red heat. This residue weight after ignition in the muffle furnace shall be multiplied by 2.5 and the product defined as the weight of the residue after ignition.

4.4.2.2.3 *Acid or alkali.*—A 400 cc. of the remaining water extract (see 4.4.2.2) shall be divided into two 200 cc. portions. Three drops of phenolphthalein (T.S.) shall be added to one portion, and one drop of methyl orange (T.S.) shall be added to the other portion. The color of the water extract portions shall be noted.

4.4.2.2.4 *Starch.*—To the remaining 200 cc. (see 4.4.2.2) of water extract add one drop of iodine T.S., and the color of the water extract shall be noted.

4.4.2.3 *Fatty matter.*—A 10-g. (plus or minus 10 mg.) specimen of nonwoven fabric shall be packed in a Soxhlet extractor and extracted with ether, the temperature being adjusted so that the ether will siphon over not less than four times per hour in the tared flask of the extraction apparatus. The extraction shall be continued for 5 hours. The ether extract in the flask shall show no trace of blue, green, or brownish color. The extract shall be evaporated to dryness, dried to constant weight of 105° C., and the residue weight determined.

4.4.2.4 *Ash content.*—A 5-g. specimen of nonwoven fabric, accurately weighed, shall be placed in a porcelain or platinum dish, and moistened with diluted sulfuric acid. The mixture shall be heated gently until it is charred and then more strongly ignited until the carbon is consumed. The weight of the residue shall then be determined.

4.4.2.5 *Uncombined dyes.*—A 10-g. specimen of nonwoven fabric shall be packed in a narrow percolator, and extracted slowly with alcohol until the percolate measures 50 cc. The color of the percolate shall be determined by observing the percolate downward in a column 20 cm. in depth.

4.4.3 *Gauze.* — Gauze shall be tested for compliance with Federal Specification CCC–G–101.

4.4.4 *Glassine paper.*—The weight of the glassine paper shall be determined as described in Federal Specification UU–P–31.

4.4.5 *Pad weight.*—Pad weight shall be determined on pads which have been conditioned as specified in Federal Specification CCC–T–191. Pads shall be weighed on a balance which is accurate to 0.01 gram. Pads shall be weighed individually.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102.

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5.1 For civil agencies.

5.1.1 *Packaging and packing.*—The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.1.2 *Marking.*

5.1.2.1 *Interior packages.*—Each package shall be marked to identify contents.

5.1.2.2 *Shipping containers.* — Shipping containers shall be marked with stock number, name of material, quantity, name of consignee, name of consignor, and contract number as shown on contract or purchase order. In addition to any special markings, containers shall be marked in accordance with Federal Standard No. 123.

5.2 For military agencies.

5.2.1 *Packaging.*

5.2.1.1 *Unit of issue.*—Fifty dressings, as specified, constitute one unit.

5.2.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal stock number	Unit package	Intermediate package	Exterior container
6510–299–8341	1 unit	24 units	48 units

5.2.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.1.3 *Level A.*

5.2.1.3.1 *Unit package.* — Fifty dressings packaged as specified in 3.3, shall be further packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.2.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, class 2, compliance symbol W5c or W5s, style RSC. Closure shall be effected by means of water-resistant gummed or pressure-sensitive tape. The tape shall be not less than 2 inches wide. Three strips shall be applied to both top and bottom so that all seams are covered their full length. Tape applied over the lengthwise seams shall extend at least 2 inches onto each of the end panels. Tape applied over the end-edge seams shall extend a minimum of 2 inches along the side and top, and side and bottom panels. Tape will not be required on manufacturer's joint.

5.2.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2.2 *Packing.*

5.2.2.1 *Level A.*—Exterior container shall be a box of appropriate size constructed in accordance with Federal Specification PPP–B–636, class 2, compliance symbol V3c or V3s, style RSC. Closure shall be as specified in 30.2.2.1 thereof, and, in addition, the top and bottom center seams shall be closed by means of tape conforming to the requirements of Federal Specifications PPP–T–0045 or PPP–T–76. The tape shall be not less than

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3 inches wide and shall extend at least 2 inches onto each of the end panels. Strapping will not be required.

5.2.2.2 *Level C.*—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.3 *Marking.*

5.2.3.1 *Individual package.* — Each individual sterilized package shall be commercially identified, and shall bear name of manufacturer and a statement that sterility is not guaranteed if wrapper is broken.

5.2.3.2 *Unit packages.*—Each unit package shall be commercially identified. Lot (control) number shall be shown.

5.2.3.3 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129. Lot (control) number shall be shown.

5.2.3.4 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2(c), 6.2(d), and 6.5 thereof shall not be applicable. Lot (control) numbers shall be shown. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required (see 1.2.1).

(*c*) Selection of applicable levels of packaging and packing for military procurement (see 5.2.1 and 5.2.2).

6.2 This specification covers the following item appearing in the Armed Services Medical Stock List:

6510–299–8341 PAD, COTTON, Gauze Covered, 2½ by 2⅛ inches, 50s.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 Transportation description.—Transportation description applicable to this item is:

Dressings, Surgeons'

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor volume minimum weight 24,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army — M
Navy — Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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